

Project Milestones

1. Proposal and Motivation (Weight: 10 marks)
5-minute presentations on 12 Feb 2026
2. Empirical Comparisons of Three GNNs (Weight: 10 marks)
3. Additional Insights (Weight: 10 marks)
10-15 minute presentations on 21 April 2026

Proposal, Motivation (Weight: 10 marks)

1. Group & Data: Introduce team members and select **three real-world datasets** (e.g., from [torch_geometric.datasets](#)) other than Cora, Citeseer, Pubmed
2. Dataset Specs: Detail the semantics of **nodes, edges, and node features** for all three datasets
3. Task Definition: Define the **graph neural network task** (e.g., node classification)
4. Explicitly answer:
 - a. Why are both **node features and structural edges necessary** for this specific task?
 - b. How do the node features **complement** the graph structure?

5-minute presentations on 12 Feb 2026

Methodology and Empirical Comparison (Weight: 10 marks)

1. Model Selection: Select **three distinct GNN** architectures suitable for the task defined in Milestone 1 (e.g., from <https://pytorch-geometric.readthedocs.io/en/latest/modules/nn.html>)
2. Experiments: Run **all 3 GNNs on all 3 datasets**.
3. Optimization: **Tune hyperparameters** for the best configuration for each model/dataset pair.
4. Benchmarking: Compare performance across datasets using a **primary metric** (e.g., Accuracy, F1-Score).

10-15 minute presentations on 21 April 2026

Insights and Analysis (Weight: 5 + 5 marks)

Any two of the following:

1. Ablation Studies
2. Visualisation (e.g., TSNE)
3. Efficiency Metrics
4. Robustness Analysis
5. Error Analysis by Graph Topology
6. Model Explainability

10-15 minute presentations on 21 April 2026

Analyses on the Largest Dataset

- Select the largest of the three datasets
- Choose any two of the six analysis types
- Run those two analyses across all three GNN models on the largest dataset

1. Ablation Studies

- Systematically **disable specific model components** to quantify their **contribution**.
- Verify if **sophisticated mechanisms** (graph structure or deep layers) are actually **necessary**.
- **Isolate** sources of **overfitting** or performance **gains**.

Examples

- Replace the adjacency matrix with an Identity matrix to test if the model outperforms a standard MLP.
- Replace node features with constant vectors to test if the model can learn purely from graph topology.
- Compare performance across 2, 4, and 8 layers to identify "oversmoothing" issues.

2. Visualisation

- **Project** high-dimensional node representations into **2D space**.
- **Qualitatively** assess how well the model separates different classes.
- Visually identify distinct **clusters** or confused boundaries.

Examples

- Plot 2D embeddings colored by ground-truth class to demonstrate clear cluster separation.
- Use distinct marker shapes for misclassified nodes to see if they cluster at decision boundaries.

3. Efficiency Metrics

- Evaluate the **computational** 'cost' of achieving model accuracy.
- Compare **resource consumption** across different GNN architectures.

Examples

- Specific comparison of average training time per epoch and total inference latency.
- Plot accuracy against the total count of trainable parameters for each model.
- Log and compare the peak GPU VRAM usage required for each architecture.

4. Robustness Analysis

- Test model stability against imperfect or **noisy data**.
- Measure how drastically **performance degrades** under noise.

Examples

- Plot accuracy degradation curves when randomly removing 10% to 50% of graph edges.
- Add Gaussian noise to node features to simulate sensor errors and re-evaluate accuracy.
- Compare model performance when training on severely reduced datasets (e.g., only 5% of labels).

5. Error Analysis by Graph Topology

- Correlate **prediction errors** with specific **structural properties** of the nodes.
- Determine if the model is biased toward specific graph positions (e.g., hubs).

Examples

- Group test accuracy by node degree (Low vs. High) to see if hubs are classified more accurately.
- Specific analysis of performance on nodes connected to different classes vs. same-class neighbors.
- Compare accuracy on the giant connected component versus isolated outlying subgraphs.

6. Model Explainability

- Uncover **why** the model made specific predictions.
- Identify **which neighbours or features** influenced the decision the most.
- Validate that the model is learning **meaningful structural motifs**.

Examples

- Visualize GAT attention scores as edge thickness to show which neighbors were prioritized.
- Use GNNExplainer to highlight the specific sub-graph responsible for a node's classification.
- Measure the drop in confidence when specific node features are masked or randomized.